Lauren Hyoseo Yoon

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RESEARCH

I engineer **efficient, stateful foundation models**, attacking two bottlenecks that keep today's models fragmented and forgetful:

- **Pillar I — Alignment** (*space*): force independently-formed representations into one shared, low-dimensional space.
- **Pillar II — Active inference-time learning** (*time*): let a system direct its own test-time compute, choosing what to explore and what to keep so that compute compounds.

EDUCATION

Ph.D., Computation & Neural Systems (CNS) | California Institute of Technology — 2024 – present

Advisor: Yisong Yue.

Bachelor's, Mathematics & Neuroscience (Honors) | Northwestern University — 2020 – 2024

High School Diploma | Korean Minjok Leadership Academy

Selective gifted boarding school, South Korea | International Philosophy Olympiad, National Representative (South Korea).

SELECTED RESEARCH

Pillar I · Multimodal Alignment forcing independently-formed representations into one shared space

Escaping Plato's Cave: JAM for Aligning Independently Trained Vision and Language Models (link)

PMLR / NeurIPS 2025 (UniReps)

Aligns *frozen*, independently trained vision and language backbones with small per-modality autoencoders that meet at a shared bottleneck. A novel **Spread Loss** turns the Platonic Representation Hypothesis from something you *detect* into a training *objective* — matching a fine-tuned CLIP at ∼1/3 of the FLOPs. *Google DeepMind; co-advised by Been Kim & Yisong Yue.*

Synthesizing Artificial Perception with a Wireless Transcranial Optogenetic Array (link)

Nature Neuroscience 2025

Writes spatiotemporally *patterned* percepts into cortex via an addressable micro-LED array and **decodes them to behavioral choice**; discrimination scales with stimulation-site distance (PCA-denoised). Installing a representation and reading out what the animal made of it.

Striatal Ensemble Activity in an Innate Naturalistic Behavior (link)

eLife 2023

A streaming **perception-to-representation pipeline**: lifts markerless video to 3D pose, then aligns it with unsupervised **population-level** neural trajectories (NMF) — movement and neural dynamics in one shared low-dimensional space. The same alignment problem as the AI work, in biology.

Wireless Multilateral Optofluidic Microsystems for Programmable Optogenetics & Photopharmacology (link)

Nature Communications 2022

An **online connector system** for the brain — a GUI front-end bound to a high-frequency biological-signal backend over an NFC/RF link, driving up to 256 battery-free implants in real time, with behavior read back from video. The systems-engineering instinct that carries forward.

Pillar II · Active Inference-Time Learning directing test-time compute — what to explore, what to keep

Knowledge-Centric Self-Improvement (KCSI) (link)

COLM 2026 (CBW)

Inverts agent-centric self-improvement: agents stay generic and **disposable**, and the only thing that persists is a shared **curated knowledge base**. Transient agents attempt tasks and debate across them; what survives is distilled into reusable bundles. Higher solve rate at substantially lower cost (ARC, Polyglot, SWE-Bench Pro), 47.2% on Terminal-Bench, and bundles that transfer **zero-shot** across model families. *Current direction.*

VendiEvolve: Towards Diversity-Aware LLM-Guided Evolutionary Search (link)

ICML 2026 (Human-AI Co-Creativity)

LLM-guided evolutionary search collapses onto a single program family, losing the stepping-stones to better solutions. VendiEvolve

puts the **Vendi Score** — the effective number of distinct programs — into the loop as a control signal steering selection toward diversity. **Diversity as a system control signal**; the population is the product, not just the single winner.

When Models Choose the Question: Pedagogical Constraints in Bottom-Up Multi-Agent Inquiry (link)

Preprint · arXiv 2026

What shapes a model's inquiry when no question is handed to it to begin with? A **bottom-up forum**, modeled on Philosophy for Children, where LLM agents read a philosophical narrative and propose, select, and answer their own questions, reaching a shared conclusion with no facilitator or aggregator. The aim: test whether the freedom to formulate questions genuinely produces **self-directed inquiry**, with the **ECO framework** as the lens for tracing where that inquiry draws from.

EXPERIENCE

Student Researcher | Google (Seattle) Starting Dec 2026

Student Researcher | Google DeepMind / Brain Jun 2024 – May 2025

Multimodal alignment of frozen vision–language models (JAM).

Research Intern, Biomedical Computing (Health Futures) | Microsoft Research Jun – Sep 2023

Foundation models of human DNA; proposed a genotype-imputation fine-tuning task and an efficient raw-data handling framework.

Research Scientist | Querrey Simpson Institute for Bioelectronics, Northwestern Fall 2023 – Jun 2024

End-to-end analysis pipelines for artificial-perception experiments (optogenetic/neural signals).

Research Scientist | Kozorovitskiy Lab, Northwestern Sep 2021 – Jun 2024

Senior thesis: interpretable generative models of behavior from electrophysiology + video; co-advised by Ann Kennedy (Feinberg).

Research Scientist | Computational Neuroscience Lab, Feinberg School of Medicine (Northwestern) Mar 2022 – Jan 2024

Python/ML behavior-tracking pipeline (MARS) characterizing circuit dysfunction in a Parkinson's mouse model.

Research Assistant | Social Identity & Morality Lab, New York University Jan – Sep 2021

Built an automated social-media (Twitter API) program to test the backfire effect in fake-news discernment.